

EVEN SEMESTER (Session 2024-25)

MID SEMESTER EXAMINATION

ROLL NO.

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Subject Code & Name: LW 30201 Legal Fundamentals for Engineers: Contracts, Taxation and Beyond

Program/Branch/Year: B.A.LL.B (II)/BBA.LL.B (II) 2nd Year/SEM IV

Time: 2 Hrs

Max Marks: 60

General Instructions. Read all instructions carefully.

1. Do not write anything on the question paper except your Roll No.
2. Answers should be written in clear and legible handwriting. Neatly labeled diagrams will fetch better marks.
3. Students must clearly write the question number & sub-part of question that they are attempting.
4. This question paper consists of 3 questions and all questions are compulsory. All questions carry 20 marks each.
5. Students are expected to take about 35 minutes each for Questions 1 to 3 and Remaining 15 minutes for reading the question paper and revision

Parts i) and ii) of each question are compulsory and each part carries 2 marks. Parts iii), iv) and v) carry 8 marks each and the student may attempt any 2 parts.

Q

i) What is a contract?

[BL-1] (2 marks)

A) A social agreement

B) A legally enforceable agreement

C) A moral obligation

D) An informal understanding

ii) A contract is void if its object is:

[BL-1] (2 marks)

A) Lawful

B) Against public policy

C) For business purposes

D) In writing

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iii) Explain the meaning and essential characteristics of a valid contract with suitable examples. (BL-3) (8 marks)

iv) Define offer and acceptance. What are the essential conditions for a valid offer and a valid acceptance? (BL-3) (8 marks)

v) Explain the different ways in which a contract can be discharged.

(BL-3) (8 marks)

Q2

i) Which of the following is an essential feature of a partnership?

[BL-1] (2 marks)

- a) Separate legal entity
- b) Limited liability of partners
- c) Sharing of profits and losses
- d) Issuing of shares

ii) The registration of a partnership firm under the Indian Partnership Act, 1932 is:

[BL-1] (2 marks)

- a) Compulsory
- b) Optional
- c) Mandatory for firms with more than 5 partners
- d) Mandatory for firms with foreign partners

iii) Define partnership as per the Indian Partnership Act, 1932. What are its essential features?

(BL-3) (8 marks)

iv) Discuss the duties and rights of partners in a partnership firm as per the Indian Partnership Act, 1932.

(BL-3) (8 marks)

v) Discuss the key features of an LLP under the LLP Act, 2008.

(BL-3) (8 marks)

Q3

i) Which right ensures protection against hazardous goods and services? [BL-1] (2 marks)

- a) Right to Information
- b) Right to Safety
- c) Right to Choose
- d) Right to Redressal

ii)) Which consumer right provides access to accurate product details? [BL-1] (2 marks)

- a) Right to be Heard
- b) Right to be Informed
- c) Right to Choose
- d) Right to Consumer Education

iii) Explain the six consumer rights recognized under the Consumer Protection Act, 2019. (BL-3) (8 marks)

iv) Explain the powers and functions of the National Consumer Disputes Redressal Commission.

(BL-3) (8 marks)

v) Explain the role of consumer education in protecting consumer rights. (BL-3) (8 marks)

EVEN SEMESTER (Session 2024-25)
MID SEMESTER EXAMINATION

ROLL NO.

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Subject Code & Name: CS34210 & Introduction to Software Engineering
Program/Branch/Year: B.Tech. IInd Year/ SEM IV
Time: 2 Hrs

Max Marks: 60

General Instructions. Read all instructions carefully.

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3. Students must clearly write the question number & sub-part of question that they are attempting.
4. This question paper consists of 3 questions and all questions are compulsory. All questions carry 20 marks each.
5. Students are expected to take about 35 minutes each for Questions 1 to 3 and Remaining 15 minutes are for reading the question paper and revision.

Note: BL refers to Blooms Taxonomy Level indicating Level of Complexity of a question

Parts i) and ii) of each question are compulsory and each part carries 2 marks. Parts iii), iv) and v) carry 8 marks each and the student may attempt any 2 parts.

Q1.

- i) After the design phase the document prepared is known as Software Requirements Specification (2 marks, BL=1)
- ii) CASE stands for Computer Aided Software Engineering (2 marks, BL=2)
- iii) Define the data flow diagram. Explain its use in software development. (8 marks, BL=2)
- iv) Write short note on:
 - a) Software crisis.
 - b) Software development process.
 - c) Characteristics of software engineering.(8 marks, BL=2)
- v) Explain the Waterfall and Spiral Model in detail. (8 marks, BL=2)

Q2.

- i) In waterfall model, output of one phase is input to next phase (True/False) (2 marks, BL=1)
- ii) Selection of life cycle model is based on software requirements (2 marks, BL=2)
- iii) Differentiate the user and system requirements in detail. (8 marks, BL=2)
- iv) Explain the software requirement analysis and modeling. (8 marks, BL=2)
- v) Explain working of SEI-CMM model in detail with diagram. (8 marks, BL=2)

Q3.

- i) In size-oriented metrics, metrics are developed based on the size of the program (2 marks, BL=1)
- ii) The following is not a direct measure of SE process
 - a) Efficiency

- b) Cost (2 marks, BL=1)
- c) Effort Applied (8 marks, BL=2)
- d) All of the mentioned (8 marks, BL=2)
- iii) Explain the characteristics of a good design? Describe different types of coupling and cohesion. (8 marks, BL=2)
- iv) Define design document? Explain Structured Coding Techniques. (8 marks, BL=2)
- v) Differentiate between a real time system design and distributed system design? (8 marks, BL=2)

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ROLL NO.

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Subject Code & Name: CS34211 & Computer Network
Program/Branch/Year: B. TECH /2nd Year/IV Sem
Time: 2 Hrs

Max Marks: 60

General Instructions. Read all instructions carefully.

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 4. **This question paper consists of 3 questions and all questions are compulsory. All questions carry 20 marks each.**
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Parts i) and ii) of each question are compulsory and each part carries 2 marks. Parts iii), iv) and v) carry 8 marks each and the student may attempt any 2 parts.

Q1.

- i) Which transmission medium offers the highest data transmission speed and resistance to electromagnetic interference? (2 marks, BL 2)
 - a) Coaxial cable
 - b) Optical fiber
 - c) Twisted pair cable
 - d) Radio waves
- ii) Which of the following best describes the difference between analog and digital signals? (2marks, BL 2)
 - a) Analog signals have discrete values, while digital signals are continuous
 - b) Analog signals are continuous, while digital signals have discrete values
 - c) Analog signals use binary encoding, while digital signals use waveforms
 - d) Analog signals are used for computing, while digital signals are used for networking
- iii) Explain the characteristics of different physical transmission media. Discuss their advantages and disadvantages. (8 marks, BL2)
- iv) Differentiate between simplex, half-duplex, and full-duplex communication with real-world examples. (8 marks, BL 3)
- v) Differentiate between analog and digital signals in the context of data communication. Discuss their impact on network performance. (8 marks, BL3)



Q2.

i) In the OSI model, which layer is responsible for converting data into signals for transmission over a physical medium? (2 marks, BL2)

- a) Transport Layer
- b) Network Layer
- c) Data Link Layer
- d) Physical Layer

ii) A network administrator needs to implement an error detection mechanism that can also correct single-bit errors. Which technique should they use? (2 marks, BL2)

- a) Parity Check
- b) Cyclic Redundancy Check (CRC)
- c) Hamming Code
- d) Checksum

iii) Differentiate OSI and TCP/IP models in terms of their architecture, functions, and real-world applications. (8 marks, BL3)

iv) Differentiate error detection and correction techniques. Which method is most efficient for reliable communication? (8 marks, BL5)

v) Discuss how the sliding window protocol works in flow control and its significance in ensuring efficient data transmission. (8 marks, BL2)

Q3.

i) An organization is expanding its network and needs to divide its large IP address block into smaller subnetworks for better management. What should they use? (2 marks, BL2)

- a) Network Address Translation (NAT)
- b) Subnetting
- c) ARP Protocol
- d) VLAN

ii) A bank wants to implement a network infrastructure that ensures real-time communication between branches while minimizing data loss. Which switching technique is best suited for this purpose? (2 marks, BL2)

- a) Circuit Switching
- b) Message Switching
- c) Packet Switching
- d) Frequency Switching

iii) Differentiate between IP addressing and MAC addressing.

(8 marks, BL 3)

ii) Propose a method to improve internetworking between IPv4 and IPv6 networks. Discuss potential challenges and solutions. (8 marks, BL3)

iii) Explain a hybrid switching strategy that combines circuit switching and packet switching to optimize network performance in a telecom system. (8 marks, BL3)

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ROLL NO.

2401301902

Subject Code & Name: CS34212 & Object-Oriented Programming and System with Java
Program/Branch/Year: B.TECH 2nd Year/ SEM IV
Time: 2 Hrs

Max Marks: 60

General Instructions. Read all instructions carefully.

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Q1.

i) `int x=0, y=0, z=1;`

`x= (++x + y--)*z++;` what will be the value of x

[2 marks, BL-2].

- a) 2
- b) -1
- c) 0
- d) 1

ii) In java arrays are:

[2 marks, BL-2]

- a) Objects
- b) Objects reference.
- c) Primitive data type
- d) None of above.

iii) Define an Array. Classify the types of arrays in Java. Explain how arrays are created and declared in java? Give an example.

[8marks, BL-3]

iv) Explain the need of constructor in a class also explain the types of constructor and its properties.

[8 marks, BL-2]

v) Describe object-oriented methodology? Explain the features of OOPs in brief. Also write a Java program to validate your answer.

[8 marks, BL-3]

Q2.

i) which of these methods of class string is used to obtain length of string object.

[2 marks, BL-1]

- a) `get()`
- b) `sizeof()`
- c) `lengthof()`
- d) `length()`

ii) private member functions _____

[2 marks, BL-1]

- a) Can't be called from enclosing class
- b) Can be accessed from enclosing class
- c) Can be accessed only if nested class is private
- d) Can be accessed only if nested class is public

iii) Discuss the usage of the super keyword in Java. Explain its role in invoking superclass methods, variable and constructors.

[8 marks, BL-2]

iv) Explain the Inheritance and its type. Why multiple inheritance not supported in java? Also write a program of inheritance.

[8 marks, BL-2]

v) Differentiate between class, abstract class, and interface. Justify your answer with an example. [8 marks, BL-3]

Q3.

i) Find out the exception which is not type of the exception.

[2 marks, BL-2]

- a) Checked Exception
- b) Unchecked Exception
- c) Runtime Exception
- d) Static Exception

ii) Find out the method which is used to pause the execution of a thread for a specific amount of time. [2 marks BL-1]

- a) sleep()
- b) pause()
- c) wait()
- d) yield()

iii) Describe the usage of try, catch, throw, throws, and finally in exception handling. Provide examples to illustrate their usage.

[8 marks, BL-2]

iv) Explain the Differentiate between User Define package and inbuilt package with a suitable example. [8 marks, BL-2]

v) Discuss the concept of polymorphism and its types with suitable program.

[8 marks, BL-2]

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MID SEMESTER EXAMINATION

Even Semester 2024-2025

ROLL NO.

2901301902

Subject Code & Name: CS34213 Theory of Automata & Formal Language

Program/Branch/Year: B.TECH CSE/AIML/CSCQ/DS/FSD/CTIS/SEM IV

Time: 2 Hrs.

Max Marks: 60

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 5. Students are expected to take about 30 minutes each for Questions 1 to 3 and remaining 30 minutes are for reading the question paper and revision.
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Q1.

i) Given: $\Sigma = \{a, b\}$ $L = \{x \in \Sigma^* | x \text{ is a string combination}\}$ Σ^4 represents which among the following? [BL-2].

- a) $\{aa, ab, ba, bb\}$
- b) $\{aaaa, abab, \epsilon, abaa, aabb\}$
- c) $\{aaa, aab, aba, bbb\}$
- d) All of the mentioned

(2 marks)

ii) The Grammar can be defined as: $G = (V, \Sigma, p, S)$. In the given definition, what does S represents? [BL-1].

- a) Accepting State
- b) Starting Variable
- c) Sensitive Grammar
- d) None of these

(2 marks)

iii) Draw a DFA to accept string of 0's and 1's ending with the string 011 [BL-2].

(8 marks)

iv) Design a finite automaton which performs addition of two binary numbers also draw a DFA to accept string 0's and 1's ending with 011 [BL-2].

(8 marks)

v) Construct a deterministic finite automata (DFA) that recognizes the language $L = \{x \in (0,1) \mid (x \text{ contains at least two consecutive 0's and } x \text{ does not contain two consecutive 1's})\}$ [BL-3].

(8 marks)

Q2.

i) The solution for the equation $R = Q + RP$ is [BL-1]

- a) $R = QP^*$
- b) $R = Q^*P$
- c) $P = RQ^*$
- d) $R = P$

(2 marks)

ii) Consider the following regular expression $R = (ab + abb)^* bbab$ Which of the following is not a set denoted by R? [BL-2]

- a) abababab
- b) ababbabbbab
- c) abbbab
- d) abtabbbab.

(2 marks)

iii) Construct NFA for following regular expressions.

i. $a^* b (a + b)^*$

ii. $(aa + bb)^* bb (a + b)^*$ [BL-4]

(8 marks)

iv) State the pumping lemma theorem for regular sets. Show that the language $L = \{0^n \mid n \text{ is prime}\}$ is not regular. [BL-3].

(8 marks)

v) Show that $(a^* b^*)^* = (a + b)^*$ [BL-3]

(8 marks)

Q3.

i) Which among the following is a CFG for the given Language

$L = \{x \in \{0,1\}^* \mid \text{number of zeroes in } x = \text{number of one's in } x\}$ [BL-2].

- a) $S \rightarrow e \mid 0S1 \mid 1S0 \mid SS$
- b) $S \rightarrow 0B \mid 1A \mid e \quad A \rightarrow 0S \quad B \rightarrow 1S$
- c) All of the mentioned
- d) none of these

(2 marks)

The following move of PDA is on the basis of-[BL-2].

- a) Present space
- b) input symbol
- c) both a and b
- d) none of the mentioned

(2 marks)

iii) Construct PDA to accept by final state the language of all strings of 0's and 1's such that number of 1's is less than number of 0's. Also convert the PDA to accept by empty stack [BL-3].

(8 marks)

iv) Consider CFG with productions [BL-3].

$S \rightarrow aB/bA$

$A \rightarrow a/aS/bAA$

$B \rightarrow b/bS/aBB$ For string "aaabbabbba" Find leftmost and rightmost derivation.

(8 marks)

v) Define CNF and GNF with suitable examples [BL-2].

(8 marks)